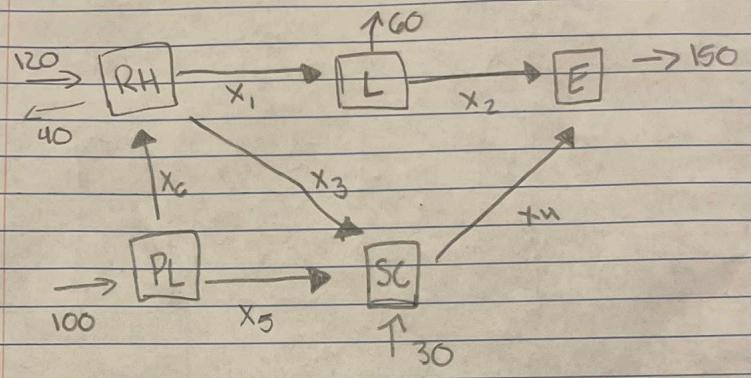
Hand write - Verification form

Shuttle → 5 stops → equation → flow in = flow out

6 routs → $X_1, X_2 \ldots$ → # bus / per hour

= Network → Linear system



① Conservation Equations:

Stop	Flow enter	Flow Exit	Conservation
Res. Hall	$120 + X_6$	$40 + X_3 + X_1$	$120 + X_6 = 40 + X_3 + X_1$
Library	X_1	$60 + X_2$	$X_1 = 60 + X_2$
Engenering	$X_2 + X_4$	150	$X_2 + X_4 = 150$
Stud. Cent	$X_3 + X_5 + 30$	X_4	$X_3 + X_5 + 30 = X_4$
Park lot	100	$X_5 + X_6$	$100 = X_5 + X_6$

Complete System:

$X_3 + X_5 + 30 = X_4$

$120 + X_6 = 40 + X_3 + X_1$

$X_1 = 60 + X_2$

$X_2 + X_4 = 150$

$100 = X_5 + X_6$.

② Augmented Matrix: 5×7 w/col. $x_1 x_2 x_3 \ldots$

$X_3 + X_5 + 30 = X_4$ → $X_3 + X_5 - X_4 = -30$

$120 + X_6 = 40 + X_3 + X_1$ → $X_6 - X_3 - X_1 = -80$

$X_1 = 60 + X_2$ → $X_1 - X_2 = +60$

$X_2 + X_4 = 150$ → $X_2 + X_4 = 150$

$100 = X_5 + X_6$ → $-X_5 - X_6 = -100$

$x_1 \; x_2 \; x_3 \; x_4 \; x_5 \; x_6$

into Matrix →

$$\left[\begin{array}{cccccc|c} 0 & 0 & 1 & -1 & 1 & 0 & -30 \\ -1 & 0 & -1 & 0 & 0 & 1 & -80 \\ 1 & -1 & 0 & 0 & 0 & 0 & 60 \\ 0 & 1 & 0 & 1 & 0 & 0 & 150 \\ 0 & 0 & 0 & 0 & 1 & 1 & 100 \end{array}\right]$$

③ Row Reduction: Gauss-Jordan → RREF

$R_1 \leftrightarrow R_3$

$$\left[\begin{array}{cccccc|c} 1 & -1 & 0 & 0 & 0 & 0 & 60 \\ -1 & 0 & -1 & 0 & 0 & 1 & -80 \\ 0 & 0 & 1 & -1 & 1 & 0 & -30 \\ 0 & 1 & 0 & 1 & 0 & 0 & 150 \\ 0 & 0 & 0 & 0 & 1 & 1 & 100 \end{array}\right]$$

$R_2 \to R_2 + R_1$, $R_3 \leftrightarrow R_4$ →

$$\left[\begin{array}{cccccc|c} 1 & -1 & 0 & 0 & 0 & 0 & 60 \\ 0 & -1 & -1 & 0 & 0 & 1 & -20 \\ 0 & 1 & 0 & 1 & 0 & 0 & 150 \\ 0 & 0 & 1 & -1 & 1 & 0 & -30 \\ 0 & 0 & 0 & 0 & 1 & 1 & 100 \end{array}\right]$$

$R_2 \leftrightarrow R_3$ →

$R_3 \to R_3 + R_2$

$$\left[\begin{array}{cccccc|c} 1 & -1 & 0 & 0 & 0 & 0 & 60 \\ 0 & 1 & 0 & 1 & 0 & 0 & 150 \\ 0 & 0 & -1 & 1 & 0 & 1 & 130 \\ 0 & 0 & 1 & -1 & 1 & 0 & -30 \\ 0 & 0 & 0 & 0 & 1 & 1 & 100 \end{array}\right]$$

$R_3 \to -R_3$, $R_4 \to R_4 - R_3$ →

$$\left[\begin{array}{cccccc|c} 1 & -1 & 0 & 0 & 0 & 0 & 60 \\ 0 & 1 & 0 & 1 & 0 & 0 & 150 \\ 0 & 0 & 1 & -1 & 0 & -1 & -130 \\ 0 & 0 & 0 & 0 & 1 & 1 & 100 \\ 0 & 0 & 0 & 0 & 1 & 1 & 100 \end{array}\right]$$

$x_1 \; x_2 \; x_3 \; x_4 \; x_5 \; x_6$

$R_5 \to R_5 - R_4$, $R_1 \to R_1 + R_2$ →

$$\left[\begin{array}{cccccc|c} 1 & 0 & 0 & 1 & 0 & 0 & 210 \\ 0 & 1 & 0 & 1 & 0 & 0 & 150 \\ 0 & 0 & 1 & -1 & 0 & -1 & -130 \\ 0 & 0 & 0 & 0 & 1 & 1 & 100 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right]$$

Free Var in col 4, 6

Pivits in col. 1, 2, 3, 5

X_4 = Choose

X_6 = choose

Pivots = X_1, X_2, X_3, X_5

General Solution:

1 0 0 1 0 0	210	$X_1 + X_4 = 210$	$\rightarrow X_1 = 210 - X_4$
0 1 0 1 0 0	150	$\rightarrow X_2 + X_4 = 150$	$\rightarrow X_2 = 150 - X_4$
0 0 1 -1 0 -1	-130	$\rightarrow X_3 - X_4 - X_6 = -130$	$\rightarrow X_3 = -130 + X_4 + X_6$
0 0 0 0 1 1	100	$\rightarrow X_5 + X_6 = 100$	
0 0 0 0 0 0	0	$X_5 = 100 - X_6$	

$X_1\ X_2\ X_3\ X_4\ X_5\ X_6$

$X_1 = 210 - X_4$	$= 100$	$X_4 + X_6 \geq 130$
$X_2 = 150 - X_4$	$= 40$	$110 + 40 = 150$
$X_3 = -130 + X_4 + X_6$	$= 20$	
$X_4 =$ Free Var	$= 110$	
$X_5 = 100 - X_6$	$= 60$	
$X_6 =$ Free Var	$= 40$	

check
$120 - 110 = 100$ ✓
$150 - 110 = 40$ ✓
$-130 + 110 + 40 = 20$ ✓
$100 - 40 = 60$ ✓

⑤ Independent flow:

Free Var:	Chosen Val:
X_4	110
X_6	40

Route	calc	flow
X_1	$210 - \overset{X_4}{110} = 100$	100
X_2	$150 - \overset{X_4}{110} = 40$	40
X_3	$-130 + \overset{X_4}{110} + \overset{X_6}{40} = 20$	20
X_4	$\overset{X_4}{110} = 110$	110
X_5	$100 - 40 = 60$	60
X_6	$40 = 40$	40

Yes it is Feasible!!

The flows are all ≥ 0 which suports the nomeg requirement.

⑥ Compare with Program

route	hand flow	Program flow	Match
X_1	100	100	yes
X_2	40	40	yes
X_3	20	20	yes
X_4	110	110	yes
X_5	60	60	yes
X_6	40	40	yes

- No notes

- at first I put the Augment matrix in wrong - just typed bad - when reentered every thing worked,